# An algebraic local action does not force thinness

Human Compression project

**Abstract**

We give three explicit involutions in Sz(8) whose parabolics have orders $14, 10, 26$. The resulting rank-three coset geometry satisfies RWPRI and the algebraic definition of local 2-transitivity used in the fixed source. Its actual panels have three elements. A short matrix identity exhibits a third chamber, and a separate literal certificate checks every panel. This refutes the $q = 8$ thinness clause of the source's Theorem 6.1; it makes no assertion about the separate $q > 8$ clause.

## 1 A small matrix certificate

Work in $F = \mathbb{F}_2[X]/(X^3 + X + 1)$. Integers from zero to seven encode coefficients in the basis $1, X, X^2$, so $2 = X$, $6 = X^2 + X$, and addition is exclusive-or. Put

$$a = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 6 & 2 & 0 & 1 \\ 2 & 0 & 1 & 0 \\ 7 & 5 & 0 & 2 \\ 7 & 7 & 2 & 6 \end{pmatrix}, \quad c = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

These are elements of the classical Suzuki matrix group. To make their membership explicit, use $w = a$, $z(t) = u(0, t)$, and

$$u(s, t) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ s & 1 & 0 & 0 \\ s^5 + t & s^4 & 1 & 0 \\ s^6 + st + t^4 & t & s & 1 \end{pmatrix}.$$

Then $b = z(2)wz(2)$ and $c = u(1, 1)wz(1)wu(1, 1)^{-1}$. The classical group model and the maximality of $D_{14}$ in Sz(8) are explicit structural premises. All the calculations below use ordinary multiplication of these matrices.

The three matrices are distinct nonidentity involutions. Direct multiplication gives

$$|ab| = 7, \qquad |ac| = 5, \qquad |bc| = 13, \qquad (ac)^2(ab)^3a = (bc)^6. \tag{1}$$

For the order statements it is enough to check the indicated prime power is the identity and the product itself is not. An independent reader performs these checks by polynomial multiplication modulo $X^3 + X + 1$. Thus (1) is a small certificate that can be checked without constructing the ambient group.

## 2 The geometry and the extra chamber

Let $G = \mathrm{Sz}(8)$ and put

$$G_0 = \langle a, b \rangle, \qquad G_1 = \langle a, c \rangle, \qquad G_2 = \langle b, c \rangle.$$

We use left cosets, in agreement with the source's convention. Two involutions whose product has odd prime order $p$ generate $D_{2p}$. Consequently the orders of our parabolics are $14, 10, 26$. The gcd of any two orders is two; since the corresponding parabolics share their indicated involution, we obtain

$$G_0 \cap G_1 = \langle a \rangle, \quad G_0 \cap G_2 = \langle b \rangle, \quad G_1 \cap G_2 = \langle c \rangle, \quad G_0 \cap G_1 \cap G_2 = 1. \tag{2}$$

The last equality follows from $a \neq b$.

**Theorem 1.** *The coset geometry $\Gamma(G; G_0, G_1, G_2)$ is RWPRI and locally 2-transitive under the literal algebraic-residue convention of L114. It is not thin and is not flag transitive.*

*Proof.* Two incident cosets have a common representative, which extends the pair to a chamber by its coset of the missing type. The same observation extends smaller flags. Hence this rank-three coset pregeometry is a geometry.

For the empty flag, $G_0 = D_{14}$ is maximal in $G$. Within each parabolic, each intersection subgroup in (2) has prime index, respectively seven, five or thirteen, and hence is maximal. The trivial triple intersection is maximal in each order-two pair intersection. These exhaust the subsets required by the source's RWPRI definition. Its algebraic rank-one residues are precisely the regular actions $C_2/1$ on two points, so are 2-transitive.

Consider the base edge $(G_1, G_2)$. The cosets $G_0$ and $cG_0$ give its two standard extensions. There is also the coset $xG_0$, where $x = (ac)^2$. It meets $G_1$ at $x$, and (1) shows that it meets $G_2$ at $(bc)^6$. It is distinct from $G_0$: an element of order five cannot belong to $G_0 \cap G_1 = \langle a \rangle$.

It is distinct from $cG_0$ as well. If they were equal, then $cx \in G_0 \cap G_1 = \langle a \rangle$. The equation $cx = 1$ is impossible since $x$ has order five and $c$ has order two. The equation $cx = a$ would give $(ac)^2 = ca = (ac)^{-1}$, contradicting $|ac| = 5$. There are therefore at least three chambers over this edge, disproving thinness. The extra chamber has no common representative: such a representative would lie in $G_1 \cap G_2 = \{1, c\}$, which would identify its third coset with one of the two standard cosets. A translate of the base chamber always has a common representative. Thus the extra chamber also disproves flag transitivity. □

## 3 Complete panels and exact source scope

The retained finite data contain the full lists of the three small parabolics. The reader verifies every group product, inverse and generated element, independently of the acquisition code. For missing type $i$, every coset incident with $G_j$ has a representative in $G_j$. Listing the distinct $gG_i$, $g \in G_j$, and retaining those meeting $G_k$ therefore checks the entire actual panel. In this example all three panels have exactly three elements, with representatives

| missing type | standard representatives | extra representative |
|---|---|---|
| 0 | $1, c$ | $(ac)^2$ |
| 1 | $1, b$ | $(ab)^3$ |
| 2 | $1, a$ | $(ba)^3$ |

Each retained coset comes with its full intersections with both base parabolics and with their intersection. Only the two standard cosets meet the latter. The short proof above needs only one extra coset; the complete panel calculation supplies the exact sizes.

The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63. Printed p. 45 defines the residue at a subflag of the distinguished chamber from intersections of parabolics. Printed p. 46 uses that residue in its local 2-transitivity condition. It separately defines thinness by the number of actual chambers over an edge and explicitly qualifies agreement with earlier definitions by flag transitivity.

The $q = 8$ clause of Theorem 6.1 on p. 61 claims that its rank-three geometries are thin. The theorem above refutes that literal clause. It is not a counterexample to the different assertion obtained by requiring flag transitivity or 2-transitivity on the full actual panels.

The discovery run stopped after the seventy-first prescribed candidate; its unmodified prefix and terminal receipt are retained. The independent reader checks this one witness, not the discovery prefix or a classification of all embeddings. The nine earlier positive markings remain valid. The rank bound and the three rank-two classes for $q = 8$ are independently accepted elsewhere. The separate thinness clause for prime $q - 1 > 7$ remains an obligation, as do the final source joins.

**Attribution and credit.** The group model, incidence definitions and maximal-subgroup premise are borrowed as stated. The explicit witness, identity (1), source comparison and independent finite reader are the contribution of this component. No priority or later-edition claim is made. This resolves one additional claim in an original whose principal refutation is already registered; it adds no original, principal-refutation, conjecture, Lean or compression count. General source application still requires distinct mathematical review.